AWS RDS Practical Assignment

DB Subnet Group • RDS Database • Endpoint Connection • Read Replica • Promotion

Objective

Create a DB subnet group and an Amazon RDS database, connect using its endpoint, create a read replica, and promote the replica to a standalone primary database. This sequence follows the supplied Rds Practical.pdf.

Prerequisites and safety

- AWS account with RDS, EC2/VPC, and security-group permissions; select one AWS Region.
- A VPC with subnets in at least two Availability Zones, and a client (such as EC2) that can reach the database.
- Use a MySQL RDS engine for this example; menu labels and replica requirements can differ by engine and AWS console version.
- RDS, storage, and EC2 may incur charges. Prefer low-cost eligible configurations, avoid public database exposure, and delete lab resources when finished.
- Keep database passwords private. Never include passwords or secrets in screenshots.

Task 1 — Create a DB subnet group

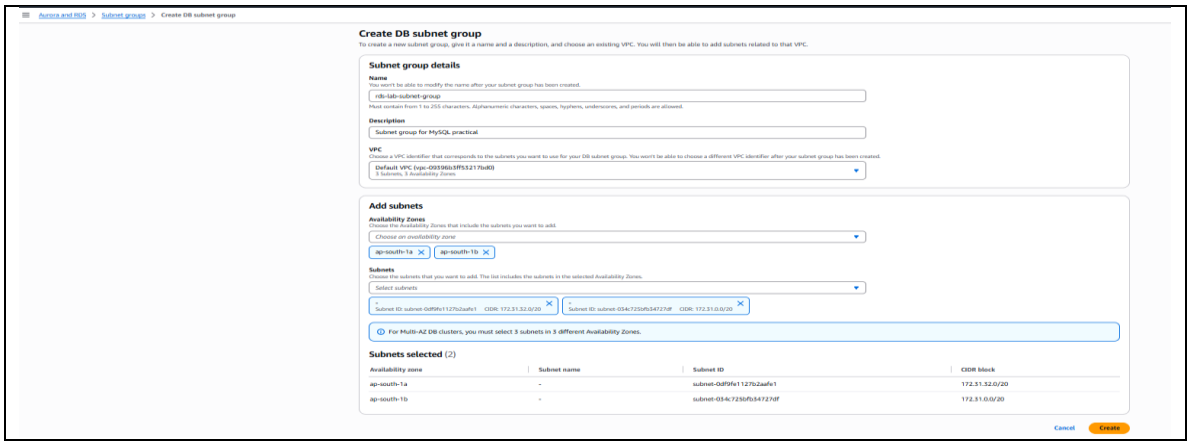
1. Open AWS Console → Amazon RDS → Subnet groups → Create DB subnet group.

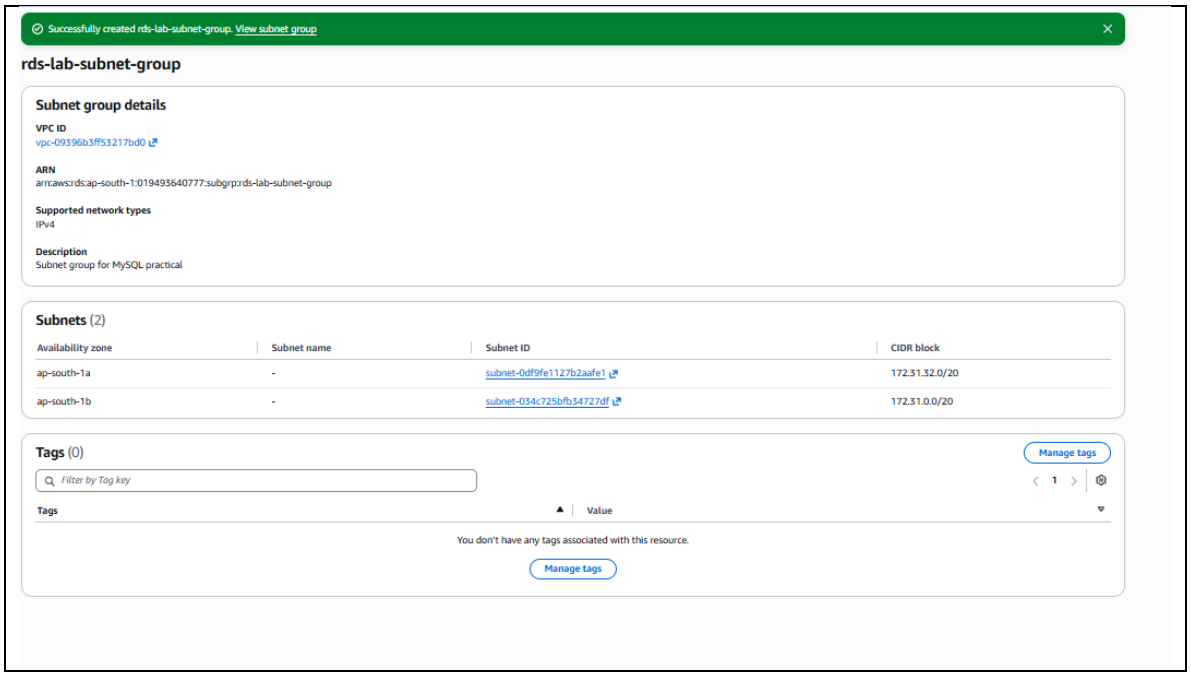
2. Enter name: rds-lab-subnet-group and a short description.

3. Choose your VPC, then select subnets from at least two different Availability Zones.

4. Choose Create and verify that the subnet group is listed.

Screenshot evidence: Subnet group details showing VPC and selected subnets





Task 2 — Create an RDS database instance

1. In Amazon RDS choose Databases → Create database → Standard create.

2. Choose MySQL engine, an available version, and a cost-appropriate template.

3. DB identifier: rds-lab-primary; set master username and a secure password (do not capture the password).

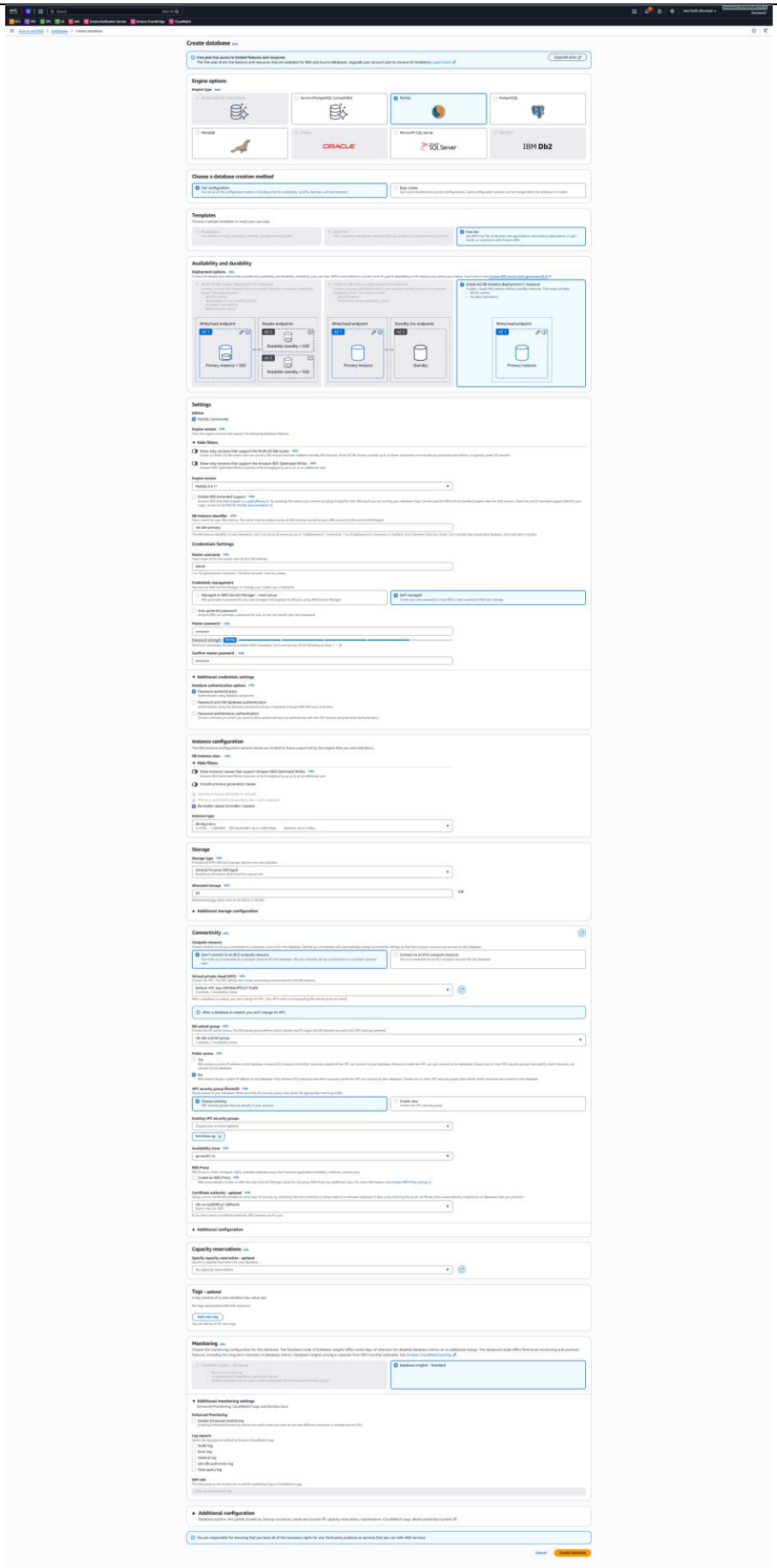
4. Choose an eligible instance class, storage, and the VPC from Task 1; select rds-lab-subnet-group.

5. Set Public access to No (recommended); select a security group that permits MySQL TCP 3306 only from your EC2/client security group.

6. Choose Create database. Wait for status Available; note the endpoint, port and DB identifier.

7. For read replicas, verify the source DB has automated backups enabled (backup retention greater than 0), if required for the selected engine.

Screenshot evidence: Database details: identifier, engine, status, connectivity (hide secrets)


Create database
Engine options
Choose a database creation method
Templates
Availability and durability
Settings
Credentials Settings
Instance configuration
Storage
Connectivity
Capacity reservations
Tags - optional
Monitoring
Additional configuration



Task 3 — Connect using the RDS endpoint

1. Use an EC2 instance or other network-reachable client with MySQL client installed.

2. Verify EC2/client and RDS network connectivity, routing and security-group permissions.

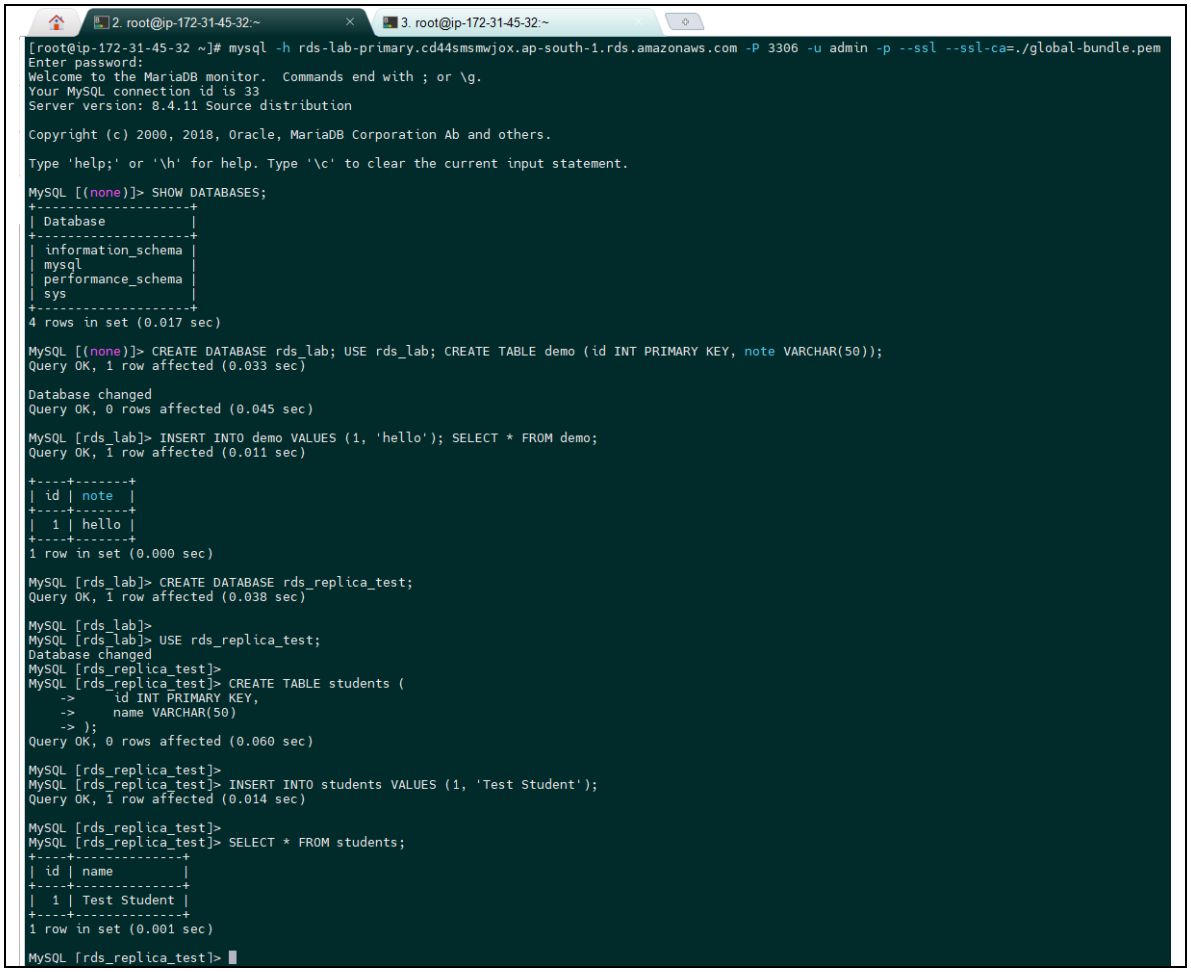
3. In RDS → Databases → rds-lab-primary → Connectivity & security, copy the endpoint and port.

4. Connect with: mysql -h <RDS_ENDPOINT> -P 3306 -u <MASTER_USERNAME> -p

5. Enter password only at the prompt; run: SELECT VERSION(); and SHOW DATABASES();

6. Optional: CREATE DATABASE rds_lab; USE rds_lab; CREATE TABLE demo (id INT PRIMARY KEY, note VARCHAR(50)); INSERT INTO demo VALUES (1, 'hello'); SELECT * FROM demo;

Screenshot evidence: Successful MySQL client connection and query output



Task 4 — Create a read replica

1. Select rds-lab-primary in Amazon RDS → Databases.

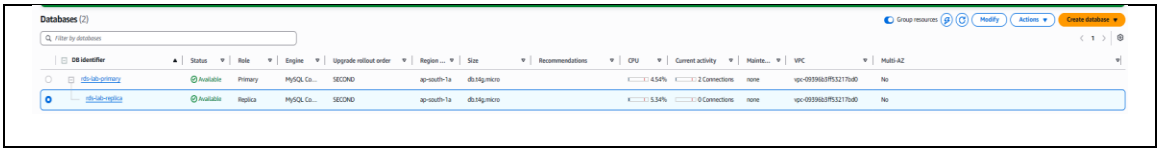
2. From Actions choose Create read replica (availability depends on engine/configuration).

3. Name the replica rds-lab-replica, choose suitable instance and network settings, then create.

4. Wait until replica status is Available. Review its endpoint and replication status.

5. Optionally connect to the replica endpoint and run SELECT * FROM rds_lab.demo; to verify replicated data. Read replicas are normally read-only.

Screenshot evidence: Replica database details and replication status




```
[root@ip-172-31-45-32 ~]# mysql -h rds-lab-replica.cd44smsmwjox.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p --ssl --ssl-ca=./global-bundle.pem
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 27
Server version: 8.4.11 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| rds_lab            |
| rds_replica_test   |
| sys                |
+--------------------+
6 rows in set (0.030 sec)

MySQL [(none)]> SELECT * FROM rds_replica_test.students;
+----+--------------+
| id | name         |
+----+--------------+
|  1 | Test Student |
+----+--------------+
1 row in set (0.003 sec)

MySQL [(none)]>
```

Task 5 — Promote read replica to standalone DB

1. Select rds-lab-replica in Amazon RDS → Databases.

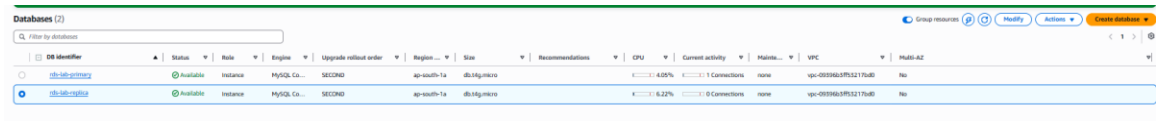
2. Choose Actions → Promote (or Promote read replica) and confirm.

3. Wait for promotion to complete; check that the replica is now a standalone DB instance and no longer replicates from the source.

4. Connect to the promoted DB endpoint and verify its data. Test writes only if appropriate for the lab.

5. Note: promotion is not automatic failover; application connections must be updated manually if switching to this DB.

Screenshot evidence: Promoted standalone DB status and endpoint




```
[root@ip-172-31-45-32 ~]# mysql -h rds-lab-replica.cd44smsmwjox.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p --ssl --ssl-ca=./global-bundle.pem
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 12
Server version: 8.4.11 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| rds_lab            |
| rds_replica_test   |
| sys                |
+--------------------+
6 rows in set (0.130 sec)

MySQL [(none)]> SELECT * FROM rds_replica_test.students;
+----+-------------+
| id | name        |
+----+-------------+
|  1 | Test Student |
+----+-------------+
1 row in set (0.069 sec)

MySQL [(none)]>
```

Observation and conclusion

Primary DB endpoint: _____

Replica endpoint: _____

Result of SELECT query: _____

Status after promotion: _____

Challenges faced / resolution: _____

Conclusion: Successfully configured an RDS database, accessed it through its endpoint, created a read replica, and promoted the replica to a standalone database.

Reference: User-provided “Rds Practical.pdf” (7 pages). The console details, MySQL commands, safety and cleanup notes above are additional implementation guidance.